

LOB Event-Stream Models — Comparison Report

Corrected 7-model benchmark + the SS2P2 head/dynamics ablation chain (Gemini-ETH)

1 Setup and validity

All models trained and evaluated identically (Gemini ETH 7d single-item, seq 64/stride 32, 40 epochs, categorical marks, hidden 64, A40). This supersedes the earlier report: three implementation bugs in the vendored baselines were found and fixed (a one-event-stale state anchor in `get_hidden_h`; the neural-Hawkes decay applied with the previous inter-event gap; detached channel-embedding gradients), and all baselines retrained. Models whose code was untouched reproduced their previous rows bit-identically (S2P2, PCT-LSTM, SS2P2-G1), validating the rerun. *hawkes* and *ct-lstm* are the same class (Mei–Eisner Neural Hawkes CT-LSTM) and are reported once as **NHP**. There is no classical parametric-Hawkes baseline in this set.

2 Prediction (per genuine test event)

overall = time + mark NLL (fit gate); mean *u* = time-rescaling calibration ($\rightarrow 1$); KS = distance of compensator masses from $\text{Exp}(1)$.

model	overall↓	timeNLL	markNLL	KS↓	mean <i>u</i>	tMAE(s)	ACC↑	PPL↓
NHP (CT-LSTM)	0.667	−1.531	2.198	0.235	1.91	0.300	0.379	9.01
SS2P2-softmin	0.982	−1.333	2.315	0.189	1.84	0.384	0.331	10.12
SS2P2-G1 (orig.)	1.078	−1.219	2.297	0.253	2.09	0.344	0.343	9.95
SS2P2-lsinit	1.149	−1.163	2.312	0.237	2.18	0.329	0.337	10.09
PCT-LSTM	1.327	−1.214	2.541	0.189	1.87	0.397	0.276	12.70
SAHP	1.524	−0.775	2.299	0.448	0.99	0.435	0.356	9.97
LSTM	1.593	−0.751	2.344	0.485	0.78	0.464	0.343	10.43
S2P2	2.959	0.613	2.346	0.437	4.29	1.430	0.326	10.45

NHP is the best predictor across the board. **SS2P2-softmin is the best of everything else** — and holds the best timing *calibration* in the entire table (KS 0.189, mean *u* 1.84, beating NHP) — while every SS2P2 variant beats its unbounded S2P2 parent by ≥ 1.8 nats.

3 Simulation — stylized-facts fit (600 s closed-loop, rel-err vs real)

model	sim rate (real 2.32)	rate_re↓	Fano_re↓	clus_re↓	retACF_re↓
SAHP	1.76	0.24	0.29	0.08	0.38
LSTM	1.20	0.49	0.46	0.28	0.23
SS2P2-G1 (orig.)	52.7	21.7	1.02	0.22	0.40
SS2P2-lsinit	78.4	32.8	0.27	1.06	0.54
SS2P2-softmin	32.3	12.9	1.97	3.69	3.52
PCT-LSTM	23.6	9.2	0.34	0.32	1.11
NHP (CT-LSTM)	66.4	27.6	6.63	3.33	0.24
S2P2	87.3	36.6	11.63	2.60	0.00

The mirror image of prediction: NHP’s properly-wired self-excitation runs away closed-loop (66 ev/s, Fano $6.6\times$ off); the bounded SS2P2 family stays contained. Softmin’s open floor lets the rollout go quiet (best rate among intensity models, 32 vs 53–87) at the cost of stronger burst–lull alternation (Fano/clustering dispersion up); the trivial LSTM/SAHP match unconditional facts best but are the weakest predictors. No model matches extreme 1s-kurtosis; all intensity models over-produce events free-running (the windowed-training/endpoint-compensator bias, mean $u > 1$ above).

4 The SS2P2 ablation chain: locating and fixing the quiet-regime gap

Per-event time-NLL deficit vs NHP, bucketed by trailing activity (last-8-gap rate; 2,957 events):

deficit vs NHP (nats)	Q1 quiet	Q2	Q3	Q4	Q5 bursts
SS2P2-G1 (sandwich head)	+1.13	+0.56	+0.37	+0.09	−0.01
SS2P2-lsinit (slow modes)	+1.16	+0.62	+0.40	+0.18	+0.04
SS2P2-softmin (open floor)	+0.68	+0.33	+0.19	+0.11	−0.02

- **Diagnosis.** SS2P2’s G1 head bounds $\lambda \in (\ell_-, \ell_+)$ *symmetrically*; ℓ_+ buys simulation stability but $\ell_- \approx 0.36$ ev/s welds the quiet floor to the burst scale. In quiet gaps the compensator bleeds $(\ell_- - \lambda^*)\Delta t \approx 0.33\Delta t$ nats — the entire Q1 deficit. In bursts SS2P2 already matched NHP.
- **Step 1 — lsinit** (S4-style log-spaced decay init, 0.02–60 s): trained dynamics retain ~ 30 s modes and optimize better (val -18.1 vs -17.0), but the metric is flat — proving the matrix was not the binding constraint.
- **Step 2 — softmin head:** $z = c - \text{softplus}(c - w^\top h - b)$, w uncapped; ceiling $s \cdot \text{softplus}(c)$ exact (thinning bound intact), floor exactly 0. Result: realized floor $0.34 \rightarrow 0.11$, the $\lambda(\delta)$ plateau at 0.36 broken (0.19 at 5 s, 0.12 at 20 s), Q1 deficit -41% , overall NLL $1.08 \rightarrow 0.98$ — while the *training* loss got worse (-12.9 vs -18.1): the old floor partly exploited the biased endpoint compensator rather than fitting the data.
- **Remaining gap** (0.20 timing + 0.12 marks): the state still converges to the frozen ZOH asymptote $x_\infty = Bu_{\text{held}}$ ($\|x\| \approx 30$ at 60 s); the zero-asymptote ablation reaches NHP-level quiet ($\lambda \approx 0.02$). A leaky hold ($u_{\text{held}}(\delta) = e^{-\rho\delta}u_{\text{held}}$) is the identified next lever.

5 Bottom line

- **Expressivity–stability frontier:** NHP takes the expressive end (best prediction, unusable simulation); S2P2 falls off both ends; **SS2P2-softmin is the current best point on the frontier** — within 0.3 nats of NHP with best-in-table calibration, and a provably bounded, non-exploding rollout.
- The ablation chain (sandwich $\rightarrow$ slow modes $\rightarrow$ open floor) is a clean mechanistic story: each step verified by the burst table and the $\lambda(\delta)$ relaxation trace, with untouched models reproducing bit-identically as controls.
- Cross-cutting: all intensity models remain rate-inflated free-running (endpoint-rule compensator + windowed training); an unbiased MC compensator is implemented (`--mc-compensator`) and untested at scale — the natural companion to the leaky-hold experiment.

Pipeline: `scripts/run_eval_all.sh` (rerunnable); variants under `experiments/{eval_all, ss2p2_lsinit, ss2p2_softmin}`; diagnostics: burst-conditional NLL + $\lambda(\delta)$ relaxation trace. Repo [honglinfu98/simulation](https://github.com/honglinfu98/simulation).